Math Kangaroo Practice

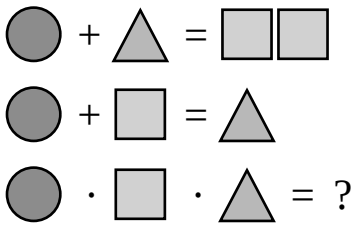
Grade 3-4 · 20 questions · standard · seed 98721

Part A: Each question is worth 3 points

1. 42 children stand in a row. Each child has a vest and a backpack. The vests repeat like this: orange, green, orange, green, ... The backpacks repeat like this: pink, brown, grey, pink, brown, grey, ... How many children have a **orange** vest **and** a **grey** backpack?

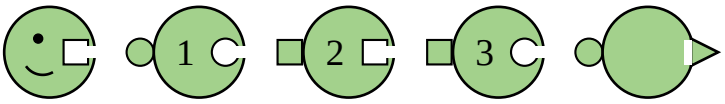
A. 3 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 35

2. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Tom lays out the cards so that the calculations are correct. What number does Tom get for the last calculation?



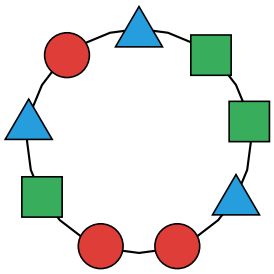
A. 0 ***B.*** 29 ***C.*** 30 ***D.*** 31 ***E.*** 60

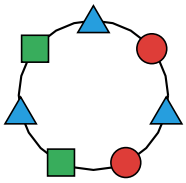
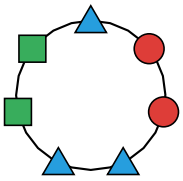
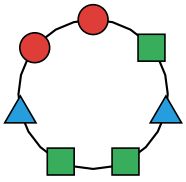
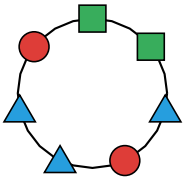
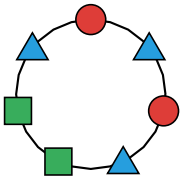
3. Mia has these pieces: a head, a tail and 3 parts with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A caterpillar is the head, then 1, 2 or 3 parts, then the tail. Each piece can be used only once. How many different caterpillars can Mia build?



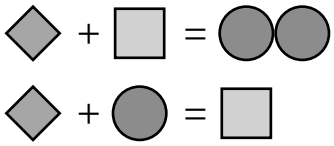
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 6

4. The picture shows a bracelet with circles (red), squares (green) and triangles (blue). Lisa removes two neighbouring stones: one square and one circle. Which bracelet can Lisa get?



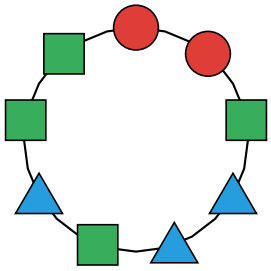
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

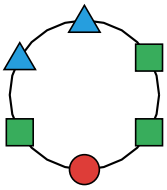
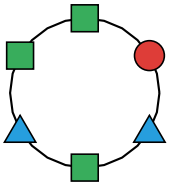
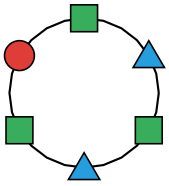
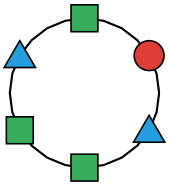
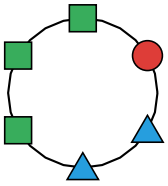
5. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Ben lays out the cards so that the calculations are correct. Which digit is hidden under the **square**?



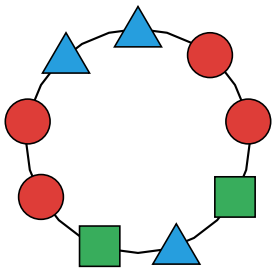
A. 1 ***B.*** 4 ***C.*** 5 ***D.*** 6 ***E.*** 7

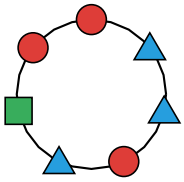
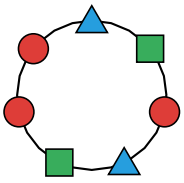
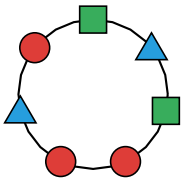
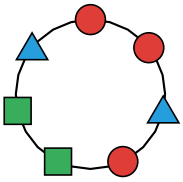
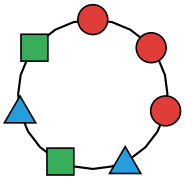
6. The picture shows a bracelet with circles (red), squares (green) and triangles (blue). Lisa removes three neighbouring stones, one of each shape. Which bracelet can Lisa get?



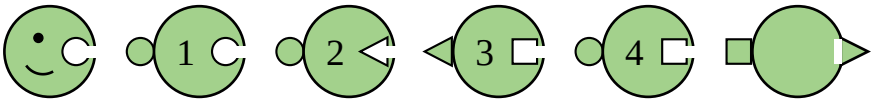
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

7. The picture shows a bracelet with circles (red), squares (green) and triangles (blue). Anna removes two neighbouring stones: one circle and one triangle. Which bracelet can Anna get?



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

8. Lucas has these pieces: a head, a tail and 4 parts with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A caterpillar is the head, then 1, 2, 3 or 4 parts, then the tail. Each piece can be used only once. How many different caterpillars can Lucas build?



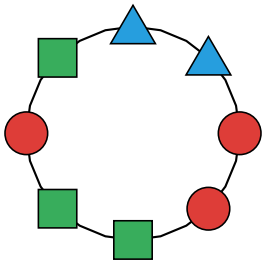
A. 3 ***B.*** 4 ***C.*** 5 ***D.*** 6 ***E.*** 7

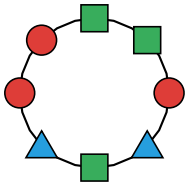
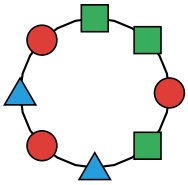
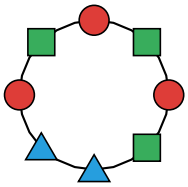
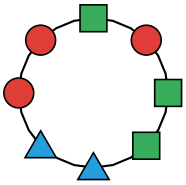
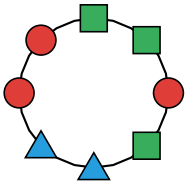
9. 30 children stand in a row. Each child has a vest and a backpack. The vests repeat like this: blue, orange, blue, orange, ... The backpacks repeat like this: pink, brown, red, pink, brown, red, ... How many children have a **blue** vest **and** a **pink** backpack?

A. 2 ***B.*** 5 ***C.*** 10 ***D.*** 15 ***E.*** 25

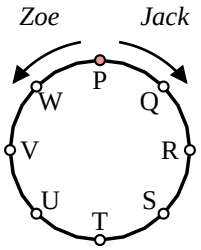
Part B: Each question is worth 4 points

10. Tom has this bracelet with circles (red), squares (green) and triangles (blue) and turns it around a bit. Which picture can show the bracelet now?



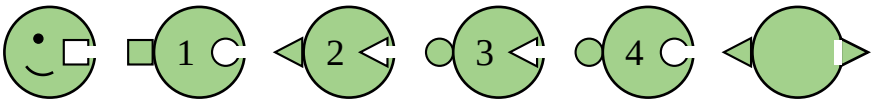
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

11. Jack and Zoe start together at point P and run around the circle in opposite directions: Jack clockwise and Zoe anticlockwise (see picture). The points are evenly spaced. Jack runs 3 times as fast as Zoe. At which point do they meet for the first time?



A. point Q ***B.*** point R ***C.*** point S ***D.*** point U ***E.*** point V

12. Ben has these pieces: a head, a tail and 4 parts with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A caterpillar is the head, then 1, 2, 3 or 4 parts, then the tail. Each piece can be used only once. How many different caterpillars with **exactly 3** parts can Ben build?

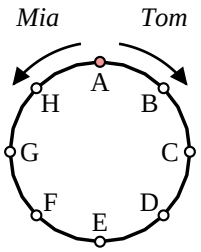


A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

13. 46 children stand in a row. Each child has a vest and a backpack. The vests repeat like this: yellow, blue, orange, yellow, blue, orange, ... The backpacks repeat like this: brown, pink, brown, pink, ... How many children have a **yellow** vest **and** a **pink** backpack?

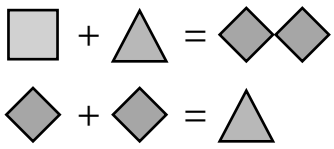
A. 3 ***B.*** 4 ***C.*** 6 ***D.*** 7 ***E.*** 8

14. Tom and Mia start at the same time from point A of the circle. Tom runs clockwise, Mia runs anticlockwise (see picture). They meet for the first time at point C, then at E, at G and then again at point A. How many times has Mia run around the circle by then?



A. 1 time ***B.*** 2 times ***C.*** 3 times ***D.*** 4 times ***E.*** 6 times

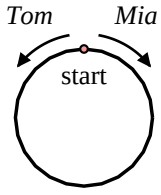
15. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Tom lays out the cards so that the calculations are correct. Which two-digit number do the cards diamond and triangle make?



A. 12 ***B.*** 13 ***C.*** 15 ***D.*** 21 ***E.*** 22

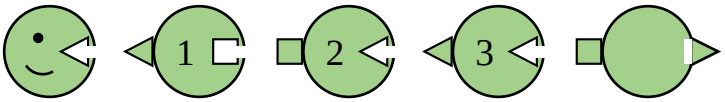
Part C: Each question is worth 5 points

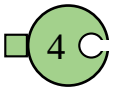
16. Mia and Tom start together at the same spot of a round track. Mia runs clockwise and Tom runs anticlockwise. Mia needs 10 minutes to run once around, Tom also needs 10 minutes. After 20 minutes they stop — both are back at the start together. How many times did they meet? (Count the meeting at the end, but not the start.)

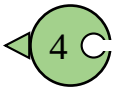


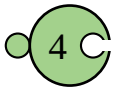
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

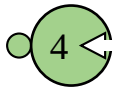
17. Lucas has these pieces: a head, a tail and 3 parts with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A caterpillar is the head, then 1, 2, 3 or 4 parts, then the tail. Each piece can be used only once. Now Lucas can build 2 caterpillars. Lucas gets one more part (number 4) and then can build exactly 4. Which part did Lucas get?

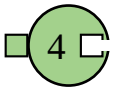


A. 

B. 

C. 

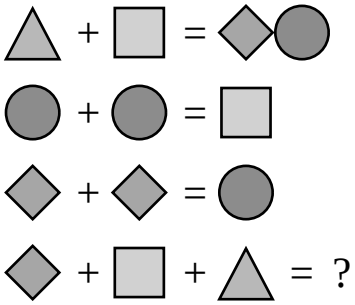
D. 

E. 

18. 24 children stand in a row. Each child has a vest and a backpack. The vests repeat like this: yellow, orange, blue, green, yellow, orange, ... The backpacks repeat like this: purple, pink, purple, pink, ... How many children have a **blue** vest **and** a **purple** backpack?

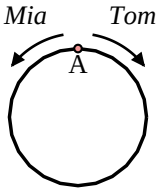
A. 0 ***B.*** 3 ***C.*** 6 ***D.*** 7 ***E.*** 12

19. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Lucas lays out the cards so that the calculations are correct. What number does Lucas get for the last calculation?



A. 11 ***B.*** 12 ***C.*** 13 ***D.*** 14 ***E.*** 15

20. Tom runs clockwise and Mia runs anticlockwise around a circle. They start together at point A. When they are together at point A again for the first time, Tom has run around the circle once and Mia 5 times. How many times did they meet? (Count the meeting at A, but not the start.)



A. 6 ***B.*** 7 ***C.*** 8 ***D.*** 9 ***E.*** 11

Answer Key

1.	**D**	**2.**	**C**	**3.**	**D**	**4.**	**E**	**5.**	**D**	**6.**	**B**
7.	**C**	**8.**	**B**	**9.**	**B**	**10.**	**E**	**11.**	**E**	**12.**	**C**
13.	**E**	**14.**	**C**	**15.**	**A**	**16.**	**D**	**17.**	**E**	**18.**	**C**
19.	**C**	**20.**	**A**								

Solutions

1. D (7) — Both patterns start again together every 6 children. In each block of 6 only child number 3 (then 9, 15, ...) has a orange vest and a grey backpack. 42 ÷ 6 = 7 blocks, so 7. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · both_count]

2. C (30) — Two digits add up to at most 9 + 9 = 18, so the square at the front of the two-digit answer must be 1. Try the possible digits and check every calculation: triangle = 6, circle = 5, square = 1. So 5 · 1 · 6 = 30. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · product_all]

3. D (4) — Start at the head and try every piece that fits, one branch at a time (H = head, T = tail): H–2–3–T, H–2–3–1–T, H–3–T, H–3–1–T. That is 4. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · count_all]

4. E — Go around the bracelet and look at every group of 2 neighbours (also the ones across the top). Taking away the group circle, square leaves the bracelet in E, turned a bit. The other pictures need stones that were not next to each other, or other stones. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · remove_two]

5. D (6) — Two digits add up to at most 9 + 9 = 18, so the circle at the front of the two-digit answer must be 1. Then try the possible digits and check both calculations: square = 6, circle = 1, diamond = 5. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · find_digit]

6. B — Go around the bracelet and look at every group of 3 neighbours (also the ones across the top). Taking away the group circle, square, triangle leaves the bracelet in B, turned a bit. The other pictures need stones that were not next to each other, or other stones. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · remove_three]

7. C — Go around the bracelet and look at every group of 2 neighbours (also the ones across the top). Taking away the group triangle, circle leaves the bracelet in C, turned a bit. The other pictures need stones that were not next to each other, or other stones. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · remove_two]

8. B (4) — Start at the head and try every piece that fits, one branch at a time (H = head, T = tail): H–1–2–3–T, H–1–4–T, H–2–3–T, H–4–T. That is 4. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · count_all]

9. B (5) — Both patterns start again together every 6 children. In each block of 6 only child number 1 (then 7, 13, ...) has a blue vest and a pink backpack. 30 ÷ 6 = 5 blocks, so 5. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · both_count]

10. E — Pick a special spot (for example a pair of equal stones) and read the stones in order from there, going around. Only E has all 8 stones in the same order; in the others two stones have swapped places. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · same_bracelet]

11. E (point V) — Until they meet they run once around the circle together, and Jack runs 3 times as far: Zoe runs 1 part and Jack 3 parts of 4. 1 part of 4 is 2 of the 8 gaps, so Zoe only gets to point V. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · where_first]

12. C (2) — Start at the head and try every piece that fits, one branch at a time (H = head, T = tail): H–1–3–2–T, H–1–4–3–T. That is 2. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · exact_length]

13. E (8) — The patterns start again together every 6 children; in each block child number 4 matches. 46 = 7 × 6 + 4: 7 full blocks give 7, and in the 4 children left over child 46 matches too. 7 + 1 = 8. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · leftover]

14. C (3 times) — Each time, Tom runs 2 of the 8 parts and Mia runs the other 6. They meet 4 times, so Mia runs 4 × 6 = 24 parts = 3 times around. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · skip_points]

15. A (12) — Two digits add up to at most 9 + 9 = 18, so the diamond at the front of the two-digit answer must be 1. Try the possible digits and check every calculation: diamond = 1, square = 9, triangle = 2. So the diamond is the tens digit and the triangle the ones digit: 12. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · plus_minus]

16. D (4) — Between two meetings they run once around the track together. In 20 minutes Mia runs 2 and Tom runs 2 times around: 2 + 2 = 4 laps together, so 4 meetings. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · minutes]

17. E — Try each new piece and count again with a tree. With E: H–1–T, H–1–4–T, H–3–1–T, H–3–1–4–T, that is 4. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · extra_piece]

18. C (6) — Write the first children down: both patterns start again together already after 4 children (not 4 × 2 = 8). In each block of 4 child number 3 matches. 24 ÷ 4 = 6 blocks: 6. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · never_together]

19. C (13) — Two digits add up to at most 9 + 9 = 18, so the diamond at the front of the two-digit answer must be 1. Start with the calculation where the fewest digits are possible, then check all three: circle = 2, square = 4, triangle = 8, diamond = 1. 1 + 4 + 8 = 13. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · four_cards]

20. A (6) — Between two meetings they run exactly once around the circle together. Together they ran 1 + 5 = 6 times around, so they met 6 times. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · count_meetings]